Ex. NO: 1

8 Queens Problem

AIM:

To write a python program to solve the 8 Queens problem using backtracking so that no two queens attack each other.

PROGRAM:

```
N = 8
def safe(b, r, c):
    return all(b[i] != c and abs(b[i]-c) != r-i
      for i in range(r))
def solve(b, r=0):
    if r == N:
        return b
    for c in range(N):
        if safe(b, r, c):
            res = solve(b + [c], r+1)
            if res:
                return res

sol = solve([])
print("Solution:", sol)
for c in sol:
    print(" ".join("Q" if i==c else "." for i in
      range(N)))
```

RESULT:

The 8 Queens Problem program was implemented in Python and executed successfully